

EFFECTIVE DISCRIMINANTS OF VERTICALLY PARAMETRIZED SYSTEMS

1. INTRODUCTION

2. MAIN RESULTS

2.1. Vertical systems. A *vertically parametrized system* in the sense of [2] is a parametric Laurent polynomial system of the form

$$C(a \star x^M) \in \mathbb{C}[a_1, \dots, a_m, x_1^\pm, \dots, x_n^\pm]^s, \quad (2.1)$$

where a is the vector of parameters, $C \in \mathbb{R}^{s \times m}$ is a full rank $s \leq n$ matrix, $M \in \mathbb{Z}^{n \times m}$ is a matrix whose columns encode the monomials and $\star$ denotes the componentwise multiplication. Motivated by applications in reaction networks theory, one is often interested in affine sections of the zero set of vertically parametrised systems. Those are modelled by *augmented vertically parametrized systems* that take the form

$$(C(a \star x^M), Lx - b) \in \mathbb{C}[a, b, x^\pm]^{s+\ell}, \quad s + \ell \leq n \quad (2.2)$$

where one adds ℓ affine equations defined by a coefficient matrix $L \in \mathbb{C}^{\ell \times n}$ and b the vector of parametric constant terms. In order to distinguish in the notation between the roles of the parameters a, b and the variables x , the specialization of the system to a choice of parameters is denoted by $F_{a,b}$ while its evaluation at a given x is denoted by $F_{a,b}(x)$.

In this work we focus on real augmented vertical systems, and more specifically, on their positive zero set. Restricting the variables of to the positive orthant allows us to generalize this parametric structure to polynomials with real exponents. Specifically, we study positive zero sets of parametric systems of the form

$$(C(a \star x^M), Lx - b) \in \mathbb{R}[a, b, x^\mathbb{R}]^{s+\ell}, \quad s + \ell \leq n \quad (2.3)$$

where now $M \in \mathbb{R}^{n \times m}$ and $\mathbb{R}[x^\mathbb{R}]$ denotes the ring of polynomials with real exponents.

An important object in our study of an augmented vertical system $F \in \mathbb{R}[a, b, x^\mathbb{R}]^{s+\ell}$ is its associated *incidence set*:

$$\mathcal{I}_{>0} := \{(a, b, x) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times \mathbb{R}_{>0}^n : F_{a,b}(x) = 0\}. \quad (2.4)$$

The following proposition adapts the structural results from [2, Theorem 3.1 and Proposition 3.2] to the setting of vertically parametrized systems with real exponents. Although the original statements were formulated for Laurent polynomial systems over $\mathbb{C}$, their proofs rely on formal differentiation and linear algebra derivations that apply verbatim to real exponent matrices $M \in \mathbb{R}^{n \times m}$, provided the domain of the variables is restricted to the positive real orthant.

Proposition 2.1. *Let $F \in \mathbb{R}[a, b, x^\mathbb{R}]^{s+\ell}$ be an augmented vertically parametrized system of the form (2.3) and let $\mathcal{I}_{>0}$ be its positive incidence variety. Then the following statements hold.*

(1) *The map*

$$\phi: \ker(C)_{>0} \times \mathbb{R}_{>0}^n \rightarrow \mathcal{I}_{>0}; \quad (w, h) \mapsto (w \star h^M, Lh^{-1}, h^{-1})$$

is a bijection, and $\mathcal{I}_{>0}$ is a smooth manifold of dimension $m + n - s$.

(2) If $\phi(w, h) = (a, b, x) \in \mathcal{I}_{>0}$, then the Jacobian matrix of $F_{a,b}$ evaluated at x satisfies

$$J_{F_{a,b}}(x) = \begin{bmatrix} C \operatorname{diag}(w) M^\top \operatorname{diag}(h) \\ L \end{bmatrix} =: Q(w, h) \in \mathbb{R}^{(s+\ell) \times n}.$$

Proof. The only part of the proof that does not follow from a direct translation of the arguments in [2, Theorem 3.1 and Proposition 3.2] is to show that $\mathcal{I}_{>0}$ is a smooth manifold of dimension $m + n - s$. To see this, consider the set

$$\mathcal{I}_{\mathbb{R}} := \{(a, b, x) \in \mathbb{R}^m \times \mathbb{R}^\ell \times \mathbb{R}_{>0}^n : F_{a,b}(x) = 0\}$$

and let $G \in \mathbb{R}^{m \times (m-s)}$ be a matrix Gale dual to C , i.e. the columns of G form a basis of $\ker(C)$. Then the map

$$\Phi: \mathbb{R}^{m-s} \times (\mathbb{R}^*)^n \rightarrow \mathcal{I}_{\mathbb{R}}; \quad (u, h) \mapsto ((Gu) \star h^M, Lh^{-1}, h^{-1}) \quad (2.5)$$

is a smooth bijection with smooth inverse that realizes $\mathcal{I}_{\mathbb{R}}$ as a smooth manifold of dimension $m + n - s$. As $\mathcal{I}_{>0} = \mathcal{I}_{\mathbb{R}} \cap \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times \mathbb{R}_{>0}^n$ is a (possibly empty) open subset of $\mathcal{I}_{\mathbb{R}}$, it inherits the smooth structure by restricting the domain of Φ to $\Phi^{-1}(\mathcal{I}_{>0})$. $\square$

2.2. Discriminating sets. Our next goal is to define subsets of $\mathcal{I}_{>0}$ whose projection into parameter space partitions it into connected components defining augmented vertical systems whose positive zero sets have the same cardinality. We will exploit the sparse structure of the set of monomials appearing in the system.

Let $\mathbf{m}_1, \dots, \mathbf{m}_m$ denote the columns of M in (2.3) and let $\mathcal{M} := \{\mathbf{m} \in \mathbb{R}^n : \mathbf{m} \text{ is a column of } M\}$. We allow M to have repeated columns, so $\#\mathcal{M} \leq m$. For Γ a face of $\operatorname{conv}(\mathcal{M})$, we define the index set of the columns that lie on this face as

$$I_\Gamma := \{j \in [m] : \mathbf{m}_j \in \Gamma\}.$$

Let $C^\Gamma \in \mathbb{R}^{s \times m}$ be the matrix obtained from $C \in \mathbb{R}^{s \times m}$ by setting all columns whose index $j \notin I_\Gamma$ to zero. For Γ a proper face of $\operatorname{conv}(\mathcal{M})$, the truncation of $\mathcal{I}_{>0}$ at Γ is defined as

$$\mathcal{I}_{>0}^\Gamma := \{(a, b, x) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times \mathbb{R}_{>0}^n : C^\Gamma(a \star x^M) = 0\}.$$

Notice that we still record the full parameter vector (a, b) even though the truncation condition drops the linear part of the original augmented system. In particular, $\mathcal{I}_{>0}^\Gamma$ is different from the incidence variety of the truncation F^Γ . For the improper face $\Gamma = \operatorname{conv}(\mathcal{M})$, we set $\mathcal{I}_{>0}^{\operatorname{conv}(\mathcal{M})} = \mathcal{I}_{>0}$. The union

$$\mathcal{F} := \bigcup_{\substack{\emptyset \neq \Gamma \text{ face} \\ \text{of } \operatorname{conv}(\mathcal{M})}} \mathcal{I}_{>0}^\Gamma \quad (2.6)$$

will be relevant in what follows, as it will be the set

$$\nabla^\circ := \{(a, b, x) \in \mathcal{I}_{>0} : \det(J_{F_{a,b}}(x)) = 0\}, \quad (2.7)$$

which detects degenerate zeros in the positive incidence variety. We will use the union

$$\nabla := \nabla^\circ \cup \left(\bigcup_{\substack{\emptyset \neq \Gamma \text{ proper face} \\ \text{of } \operatorname{conv}(\mathcal{M})}} \mathcal{I}_{>0}^\Gamma \right) \quad (2.8)$$

to partition the parameter space set into connected components for which the projection $\pi: \mathcal{F} \rightarrow \mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ has constant fiber.

2.3. Statement of the main results. We are now ready to state our main results. The following theorems establish that the number of positive real solutions of an augmented vertically parametrized system remains invariant as the parameters vary within any connected component of the complement $\pi(\nabla)$. We first present the theorem for a general exponent matrix, and then provide a refined version tailored to systems whose coefficient matrix naturally decomposes into independent blocks.

Theorem 2.2. *Let $F \in \mathbb{R}[a, b, x^{\mathbb{R}}]^n$ be an augmented vertical system of the form (2.2) with $\text{rank}(C) = s$ and $\text{rank}(L) = \ell$, $s + \ell = n$ and $\dim(\text{conv}(\mathcal{M})) = n$. Let $\pi: \mathcal{F} \rightarrow \mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ be the projection onto the first $m + \ell$ coordinates. If (a_1, b_1) and (a_2, b_2) lie in the same connected component of $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell \setminus \pi(\nabla)$, then*

$$\#V_{>0}(F_{a_1, b_1}) = \#V_{>0}(F_{a_2, b_2}).$$

Next example shows that this discriminating set may fill all the parameter space.

Example 2.3. Consider a vertical system ($\ell = 0$) with $n = 2$ variables, $s = 2$ equations, and $m = 4$ parameters, satisfying $s = n$. Let the variables be $x = (x_1, x_2) \in \mathbb{R}_{>0}^2$ and the parameters be $a = (a_1, a_2, a_3, a_4) \in \mathbb{R}_{>0}^4$. We define the exponent matrix $M \in \mathbb{R}^{2 \times 4}$ and coefficient matrix $C \in \mathbb{R}^{2 \times 4}$ as:

$$M = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 2 & 0 & 2 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & -1 & 1 \end{bmatrix}.$$

The columns of M are $\mathbf{m}_1 = (0, 0)$, $\mathbf{m}_2 = (0, 2)$, $\mathbf{m}_3 = (1, 0)$, and $\mathbf{m}_4 = (1, 2)$. The polytope $\text{conv}(\mathcal{M})$ is a rectangle in $\mathbb{R}^2$. Notice that the column index sets corresponding to the two independent sub-blocks $I_1 = \{1, 2\}$ and $I_2 = \{3, 4\}$ form the submatrices:

$$M_1 = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \quad \text{and} \quad M_2 = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}.$$

Crucially, $\Gamma_1 := \text{conv}(\mathbf{m}_1, \mathbf{m}_2)$ and $\Gamma_2 := \text{conv}(\mathbf{m}_3, \mathbf{m}_4)$ span two proper faces of $\text{conv}(\mathcal{M})$. Now, let us examine what happens under the definition of the face truncation $\mathcal{I}_{>0}^\Gamma$ for the face Γ_1 . The restricted matrix C^{Γ_1} sets to zero all columns except those in I_1 :

$$C^{\Gamma_1} = \begin{bmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

The truncated vertical system $C^{\Gamma_1}(a \star x^M) = 0$ reduces to the equation $a_1 - a_2 x_2^2 = 0$. For any choice of the parameters $a \in \mathbb{R}_{>0}^4$, one finds a positive solution of the equation by setting $x_2 = \sqrt{a_1/a_2}$ and choosing any $x_1 > 0$. Consequently, the projection of this truncated incidence variety onto the parameter space covers the entire parameter space:

$$\pi(\mathcal{I}_{>0}^{\Gamma_1}) = \mathbb{R}_{>0}^4.$$

Because the global discriminating set ∇ includes $\mathcal{I}_{>0}^{\Gamma_1}$, we get $\pi(\nabla) = \mathbb{R}_{>0}^4$.

The behaviour in [Example 2.3](#) arises because the coefficient matrix C decomposes into independent blocks, meaning its kernel naturally partitions. To address this situation, assume that, up to a simultaneous permutation of their columns, the matrices C and M can be partitioned according to r disjoint column index sets of sizes $m_1, \dots, m_r$. Specifically, C takes a block-diagonal form with blocks $C_1, \dots, C_r$ of dimensions $s_i \times m_i$, such that $\ker(C) = \ker(C_1) \times \dots \times \ker(C_r)$, and M is partitioned into column blocks $M_1, \dots, M_r$ of dimensions $n \times m_i$. This ensures that the exponent block M_i captures exactly the monomials involved in the i -th block of the kernel. Let $\mathcal{M}_1, \dots, \mathcal{M}_r$ be the corresponding sets of exponent vectors. We consider the product polytope $P := \text{conv}(\mathcal{M}_1) \times \dots \times \text{conv}(\mathcal{M}_r)$.

Let $\Gamma = \Gamma_1 \times \cdots \times \Gamma_r$ be a face of P , meaning each Γ_i is a face of $\text{conv}(\mathcal{M}_i)$. We define the truncation C^Γ as the block-diagonal matrix whose blocks are the truncated matrices $C_1^{\Gamma_1}, \dots, C_r^{\Gamma_r}$. For Γ a face of P , the truncation of $\mathcal{I}_{>0}$ at Γ is defined as

$$\mathcal{I}_{>0}^\Gamma := \{(a, b, x) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times \mathbb{R}_{>0}^n : C^\Gamma(a \star x^M) = 0\}.$$

In this partitioned setting, we define the analogues of $\mathcal{F}$ and ∇ as

$$\mathcal{F}_{\text{part}} := \bigcup_{\substack{\emptyset \neq \Gamma \text{ face} \\ \text{of } P}} \mathcal{I}_{>0}^\Gamma \quad \text{and} \quad \nabla_{\text{part}} := \nabla^\circ \cup \left(\bigcup_{\substack{\emptyset \neq \Gamma \text{ proper} \\ \text{face of } P}} \mathcal{I}_{>0}^\Gamma \right)$$

respectively.

Theorem 2.4. *Let $F \in \mathbb{R}[a, b, x^{\mathbb{R}}]^n$ be an augmented vertical system where C and M partition into r blocks as described above. Assume $\text{rank}(C) = s$, $\text{rank}(L) = \ell$, $s + \ell = n$, and that the matrix $\hat{M}$ defined in (3.1) has full row rank $n + r$. Let $\pi: \mathcal{F}_{\text{part}} \rightarrow \mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ be the projection onto the first $m + \ell$ coordinates. If (a_1, b_1) and (a_2, b_2) lie in the same connected component of $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell \setminus \pi(\nabla_{\text{part}})$, then*

$$\#V_{>0}(F_{a_1, b_1}) = \#V_{>0}(F_{a_2, b_2}).$$

Notice that Theorem 2.2 follows from Theorem 2.4 by taking $r = 1$ and $C = C_1$.

3. PROOFS OF THE MAIN RESULTS

In this section, we focus on the proof of Theorem 2.4, which is given in Section 3.3. In Section 3.1, we introduce terminology from toric geometry and in Section 3.2 we define discriminating sets in the toric setting.

3.1. Toric geometry. Given an exponent matrix $M \in \mathbb{Z}^{n \times m}$ partitioned into r blocks $M_1, \dots, M_r$ of sizes $n \times m_1, \dots, n \times m_r$, we define the partitioned matrix

$$\hat{M} := \begin{bmatrix} 1 & \cdots & 1 & 0 & \cdots & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \cdots & 0 & 1 & \cdots & 1 & \cdots & 0 & \cdots & 0 \\ \vdots & & & \vdots & & & \ddots & & & \vdots \\ 0 & \cdots & 0 & 0 & \cdots & 0 & \cdots & 1 & \cdots & 1 \\ M_1 & & & M_2 & & & \cdots & & & M_r \end{bmatrix} \quad (3.1)$$

with columns $\hat{\mathbf{m}}_1, \dots, \hat{\mathbf{m}}_m$ in $\mathbb{Z}^{n+r}$. We associate to each block the sets of vectors

$$\mathcal{M}_i := \{\mathbf{m} \in \mathbb{Z}^n : \mathbf{m} \text{ is a column of } M_i\}, \quad \text{for } i = 1, \dots, r,$$

and denote by $\hat{\mathcal{M}}$ the complete set of columns of $\hat{M}$. Note that we allow the matrices M_i to have repeated columns, so we have $\#\mathcal{M} = \#\hat{\mathcal{M}} \leq m$. The **monomial map** associated with $\hat{M}$ is the map

$$\phi_{\hat{M}}: (\mathbb{C}^*)^{n+r} \rightarrow (\mathbb{C}^*)^m; \quad t \mapsto (t^{\hat{\mathbf{m}}_1}, \dots, t^{\hat{\mathbf{m}}_m}).$$

The Zariski closure of $\text{im}(\phi_{\hat{M}})$ is known as the **affine toric variety** associated with $\hat{M}$ and denoted $Y_{\hat{M}}$, while the subsets

$$(Y_{\hat{M}})_{>0} := Y_{\hat{M}} \cap \mathbb{R}_{>0}^m \quad \text{and} \quad (Y_{\hat{M}})_{\geq 0} := Y_{\hat{M}} \cap \mathbb{R}_{\geq 0}^m$$

are called its **positive** and **nonnegative parts**.

Lemma 3.1. *Let $M \in \mathbb{Z}^{n \times m}$ be partitioned into r column blocks $M_1, \dots, M_r$, and assume that $\text{rank}(\hat{M}) = n + r$. Consider the product of probability simplices $\Delta := \Delta_{m_1-1} \times \dots \times \Delta_{m_r-1}$, where $\Delta_{m_j-1} := \{y^{(j)} \in \mathbb{R}_{>0}^{m_j} : \sum_{i=1}^{m_j} y_i^{(j)} = 1\}$. For each $x \in \mathbb{R}_{>0}^n$ and each $j \in \{1, \dots, r\}$, let $u_x^{(j)} := \frac{1}{\sum_{\mathbf{m} \in \mathcal{M}_j} x^{\mathbf{m}}}$. The map*

$$\varphi_{\hat{M}}: \mathbb{R}_{>0}^n \rightarrow (Y_{\hat{M}})_{>0} \cap \Delta; \quad x \mapsto \phi_{\hat{M}}(u_x^{(1)}, \dots, u_x^{(r)}, x)$$

is a differential isomorphism.

Proof. Since $\text{rank}(\hat{M}) = n + r$, there exists a matrix $B \in \mathbb{R}^{m \times (n+r)}$ such that $\hat{M}B = Id_{n+r}$. Let $\{b_1, \dots, b_{n+r}\}$ denote the columns of B . Then the map

$$\begin{aligned} \phi_{\hat{M}}^{-1}: (Y_{\hat{M}})_{>0} &\rightarrow \mathbb{R}_{>0}^{n+r} \\ (y_1, \dots, y_m) &\mapsto (y^{b_1}, \dots, y^{b_{n+r}}) \end{aligned}$$

is an inverse to $\phi_{\hat{M}}$. Since both of them involve only positive coordinates, both are smooth and $\phi_{\hat{M}}$ is a differential isomorphism whose restriction to the subset $\{(u_x^{(1)}, \dots, u_x^{(r)}, x) : x \in \mathbb{R}_{>0}^n\} \subseteq \mathbb{R}_{>0}^{n+r}$ gives the isomorphism in the statement. $\square$

Remark 3.2. The assumption in Theorem 2.4 that $\hat{M}$ has full row rank $n + r$ appears because in our argument we need the map $\varphi_{\hat{M}}$ in Lemma 3.1 to be a differential isomorphism. Moreover, we can reduce the case when $\text{rank}(\hat{M}) < n + r$ to the good scenario as follows. If $\text{rank}(\hat{M}) < n + r$, there is a linear dependence between the rows of M and the r block-indicator rows, meaning there exists a vector $\lambda \in \mathbb{R}^n \setminus \{0\}$ and $c_1, \dots, c_r \in \mathbb{R}$ such that $\lambda \cdot \mathbf{m} = c_j$ for all $\mathbf{m} \in \mathcal{M}_j$. Geometrically, this means that the sets of monomials $\mathcal{M}_j$ are contained in a family of parallel hyperplanes.

By performing an invertible monomial change of variables, we may assume without loss of generality that these hyperplanes are orthogonal to the n -th standard basis vector. In these new coordinates, the n -th exponent of every monomial in the block $\mathcal{M}_j$ is exactly c_j . Crucially, because the kernel decomposes into independent blocks $\ker(C) = \ker(C_1) \times \dots \times \ker(C_r)$, the incidence conditions can be evaluated block by block. For each $j \in \{1, \dots, r\}$, we have

$$C_j(a^{(j)} \star x^{M_j}) = C_j(x_n^{c_j} (a^{(j)} \star x^{M'_j})) = x_n^{c_j} C_j(a^{(j)} \star x^{M'_j}) = 0,$$

where $M'_j \in \mathbb{Z}^{(n-1) \times m_j}$ is the matrix obtained by dropping the n -th row from M_j . Since we x to the positive orthant, this is equivalent to $C_j(a^{(j)} \star x^{M'_j}) = 0$.

Therefore, the system is inherently $(n - 1)$ -dimensional in the variables, and one can redefine the ambient space by factoring out this symmetry, yielding a reduced partitioned matrix $\hat{M}'$ that successfully achieves the correct full row rank $(n - 1) + r$.

The nonnegative part of the affine toric variety $(Y_{\hat{M}})_{\geq 0}$ admits the following decomposition in terms of the faces of $\text{conv}(\hat{\mathcal{M}})$ (see [3, Equation (4)]). For each $y \in (Y_{\hat{M}})_{\geq 0} \setminus \{0\}$ there exists a nonempty face $\Gamma \subseteq \text{conv}(\hat{\mathcal{M}})$ such that

$$y_{\hat{\mathbf{m}}} > 0 \quad \text{for} \quad \hat{\mathbf{m}} \in \Gamma \cap \hat{\mathcal{M}} \quad \text{and} \quad y_{\hat{\mathbf{m}}} = 0 \quad \text{otherwise.} \quad (3.2)$$

Due to the block structure of the first r rows of $\hat{M}$, there is a natural bijection between the faces of $\text{conv}(\hat{\mathcal{M}})$ and the faces of the product polytope $P = \text{conv}(\mathcal{M}_1) \times \dots \times \text{conv}(\mathcal{M}_r)$.

Lemma 3.3. *Let $C(a \star x^M)$ define a vertical system with $a \in \mathbb{R}_{\geq 0}^m$ and C and M partitioned into r blocks as described above. Given a sequence $\{\alpha_k\}_{k \in \mathbb{N}}$ converging to a and a sequence $\{y_k\}_{k \in \mathbb{N}}$ in $(Y_{\hat{M}})_{>0} \cap \Delta$ such that $C(\alpha_k \star y_k) = 0$ for all $k \in \mathbb{N}$, there exists a subsequence of $\{y_k\}_{k \in \mathbb{N}}$ converging to some $y \in (Y_{\hat{M}})_{\geq 0} \cap \Delta$ such that $C(a \star y) = 0$.*

Proof. As the product of simplices Δ is compact, the set $(Y_{\hat{M}})_{\geq 0} \cap \Delta$ is a closed subset of a compact space, and hence compact. Thus, there exists a converging subsequence of $\{y_k\}_{k \in \mathbb{N}}$, denoted $\{y_{i_k}\}$, with limit $y \in (Y_{\hat{M}})_{\geq 0} \cap \Delta$. Using that $0 = C(\alpha_{i_k} \star y_{i_k}) = C \operatorname{diag}(y_{i_k}) \alpha_{i_k}$, we have

$$\begin{aligned} 0 \leq \|C(a \star y)\| &= \|C \operatorname{diag}(y)a\| = \|C \operatorname{diag}(y)a - C \operatorname{diag}(y_{i_k})a + C \operatorname{diag}(y_{i_k})a - C \operatorname{diag}(y_{i_k})\alpha_{i_k}\| \\ &\leq \|C \operatorname{diag}(y - y_{i_k})\| \|a\| + \|C \operatorname{diag}(y_{i_k})\| \|a - \alpha_{i_k}\|. \end{aligned}$$

As both summands converge to zero, it must hold that $C(a \star y) = 0$. $\square$

3.2. Toric discriminating sets. We define the *toric positive incidence set* in this partitioned setting as

$$\hat{\mathcal{I}}_{>0} := \left\{ (a, b, y) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{>0} \cap \Delta : (C(a \star y), L\varphi_{\hat{M}}^{-1}(y)) = (0, b) \right\}.$$

Provided that $\operatorname{rank}(\hat{M}) = n + r$, the map $\varphi_{\hat{M}}$ from Lemma 3.1 acting on the third component of $\mathcal{I}_{>0}$ defines a smooth isomorphism

$$\nu: \mathcal{I}_{>0} \longrightarrow \hat{\mathcal{I}}_{>0}; \quad (a, b, x) \mapsto (a, b, \varphi_{\hat{M}}(x))$$

and induces a commutative diagram of smooth manifolds

$$\begin{array}{ccc} \mathcal{I}_{>0} & \xrightarrow{\nu} & \hat{\mathcal{I}}_{>0} \\ \pi \downarrow & & \downarrow \tau \\ \mathbb{R}_{>0}^m \times \mathbb{R}^\ell & \xrightarrow{\operatorname{id}} & \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \end{array} \quad (3.3)$$

where both π and τ are projections onto the first $m + \ell$ coordinates.

Remark 3.4. The map Φ in (2.5) inducing the manifold structure on $\mathcal{I}_{>0}$ allows us to compute the set of critical points of π as a map between smooth manifolds. By definition, $\Phi(u, h) = (a, b, x) \in \mathcal{I}_{>0}$ is a critical point of π if

$$\operatorname{rank}(J_{\pi \circ \Phi}(u, h)) < m + \ell.$$

Following [2, Proposition 3.3], this happens if and only $\operatorname{rank}(Q(Gu, h)) < s + \ell$. In words, $\Phi(u, h) = (a, b, x)$ is a critical point of π if and only if x is a degenerate zero of $F_{a,b}$. Moreover, since the map ν from (3.3) is an isomorphism of smooth manifolds and the diagram commutes, ν specializes to a bijection between the critical points of π and those of τ .

The partitioned toric analogues $\nabla^\circ, \mathcal{I}_{>0}^\Gamma, \nabla_{\text{part}}$ and $\mathcal{F}_{\text{part}}$ are

$$\hat{\nabla}^\circ := \{(a, b, y) \in \hat{\mathcal{I}}_{>0} : \det(J_{F_{a,b}}(\varphi_{\hat{M}}^{-1}(y))) = 0\}, \quad (3.4)$$

$$\hat{\mathcal{I}}_{>0}^\Gamma := \{(a, b, y) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{\geq 0} \cap \Delta : C^\Gamma(a \star y) = 0\}, \quad (3.5)$$

$$\hat{\nabla} := \hat{\nabla}^\circ \cup \left(\bigcup_{\substack{\emptyset \neq \Gamma \text{ proper face} \\ \text{of } P}} \hat{\mathcal{I}}_{>0}^\Gamma \right) \quad \text{and} \quad \hat{\mathcal{F}} := \bigcup_{\substack{\emptyset \neq \Gamma \text{ face} \\ \text{of } P}} \hat{\mathcal{I}}_{>0}^\Gamma \quad (3.6)$$

respectively. From now on, we extend the domain of the projections π and τ from (3.3) to $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times \mathbb{R}_{>0}^n$ and to $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{\geq 0} \cap \Delta$.

Proposition 3.5. *Let $F \in \mathbb{R}[a, b, x^\pm]^n$ be an augmented vertical system where C and M partition into r blocks as described above, and assume $\operatorname{rank}(\hat{M}) = n + r$. The following statements hold.*

- (i) *Given Γ a face of P , the limit $(a, b, y) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{\geq 0} \cap \Delta$ of a converging sequence $\{(a_k, b_k, y_k)\}_{k \in \mathbb{N}} \subseteq \hat{\mathcal{I}}_{>0}^\Gamma$ satisfies $(a, b, y) \in \hat{\mathcal{I}}_{>0}^{\Gamma'}$ for Γ' a face of Γ .*
- (ii) *The set $\hat{\mathcal{F}}$ is closed in $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{\geq 0} \cap \Delta$ and contains the closure of $\hat{\mathcal{I}}_{>0}$ in the Euclidean topology.*

- (iii) For $(a, b, x) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times \mathbb{R}_{>0}^n$, it holds that $(a, b, x) \in \mathcal{F}_{\text{part}}$ if and only if $(a, b, \varphi_{\hat{M}}(x)) \in \hat{\mathcal{F}}$.
- (iv) The sets $\pi(\mathcal{F}_{\text{part}})$ and $\tau(\hat{\mathcal{F}})$ are equal and closed in $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell$.

Proof. For (i), it holds that $C^\Gamma(a_k \star y_k) = 0$ for all $k \in \mathbb{N}$ by definition of $\hat{\mathcal{I}}_{>0}^\Gamma$. Applying Lemma 3.3 to the vertical system defined by the block-diagonal matrix C^Γ guarantees that the limit satisfies $C^\Gamma(a \star y) = 0$.

By the decomposition in (3.2), we get that there exists a unique face Γ' of Γ such that $y_k > 0$ for $k \in I_{\Gamma'}$ and $y_k = 0$ for $k \in I_\Gamma \setminus I_{\Gamma'}$. Because the matrix $C^{\Gamma'}$ matches C^Γ on the columns indexed by $I_{\Gamma'}$, the action of both matrices on $a \star y$ is identical, i.e.

$$C^{\Gamma'}(a \star y) = C^\Gamma(a \star y) = 0,$$

so we conclude that $(a, b, y) \in \hat{\mathcal{I}}_{>0}^{\Gamma'}$.

To see statement (ii), consider a sequence $\{(a_k, b_k, y_k)\}_{k \in \mathbb{N}} \subseteq \hat{\mathcal{F}}$ converging to (a, b, y) in $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{\geq 0} \cap \Delta$. We establish that $\hat{\mathcal{F}}$ is closed by showing that $(a, b, y) \in \hat{\mathcal{F}}$.

Because $\hat{\mathcal{F}}$ is the finite union of toric truncations over all faces of P , there exists a face Γ such that $\hat{\mathcal{I}}_{>0}^\Gamma$ contains a subsequence. By restricting our attention to this subsequence, we may assume without loss of generality that $\{(a_k, b_k, y_k)\}_{k \in \mathbb{N}} \subseteq \hat{\mathcal{I}}_{>0}^\Gamma$. But then by (i) the limit of the subsequence lies in $\hat{\mathcal{I}}_{>0}^{\Gamma'} \subseteq \hat{\mathcal{F}}$ for Γ' a face of Γ , and hence so does (a, b, y) . This proves that $\hat{\mathcal{F}}$ is closed and since it contains $\hat{\mathcal{I}}_{>0}$, it also contains its closure.

The equivalence in (iii) holds by definition of $\hat{\mathcal{I}}_{>0}$ if $(a, b, x) \in \mathcal{I}_{>0}$. If $(a, b, x) \in \mathcal{I}_{>0}^\Gamma$ for a proper face $\Gamma = \Gamma_1 \times \cdots \times \Gamma_r$, then it is a consequence of evaluating the block-diagonal structure block by block:

$$\begin{aligned} (a, b, x) \in \mathcal{I}_{>0}^\Gamma &\Leftrightarrow C_j^{\Gamma_j}(a^{(j)} \star x^{M_j}) = 0 \quad \forall j \in [r] \\ &\Leftrightarrow u_x^{(j)} C_j^{\Gamma_j}(a^{(j)} \star x^{M_j}) = 0 \quad \forall j \in [r] \\ &\Leftrightarrow C^\Gamma(a \star \varphi_{\hat{M}}(x)) = 0. \end{aligned}$$

For the equality of sets in (iv), the inclusion $\pi(\mathcal{F}) \subseteq \tau(\hat{\mathcal{F}})$ follows directly from (iii). For the other inclusion, take $(a, b) \in \tau(\hat{\mathcal{F}})$. This means there exists $y \in (Y_{\hat{M}})_{\geq 0} \cap \Delta$ such that $C^\Gamma(a \star y) = 0$ for some Γ face of P . By the face decomposition from (3.2), there exists $\Gamma' = \Gamma'_1 \times \cdots \times \Gamma'_r$ a face of Γ , $x \in \mathbb{R}^n$, and constants $c_1, \dots, c_r \in \mathbb{R}_{>0}$ such that for each block j , $y_k^{(j)} = c_j x^{m_k}$ if $k \in I_{\Gamma'_j}$ and $y_k^{(j)} = 0$ otherwise. This implies that for each block j ,

$$0 = C_j^{\Gamma'_j}(a^{(j)} \star y^{(j)}) = c_j C_j^{\Gamma'_j}(a^{(j)} \star x^{M_j}),$$

so $C^{\Gamma'}(a \star x^M) = 0$, which implies $(a, b) \in \mathcal{I}_{>0}^{\Gamma'} \subseteq \pi(\mathcal{F})$. Hence, the equality of sets holds. Moreover, $\tau(\hat{\mathcal{F}})$ is closed due to the projection $\tau: \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{\geq 0} \cap \Delta \rightarrow \mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ being a closed map because $(Y_{\hat{M}})_{\geq 0} \cap \Delta$ is compact and $\hat{\mathcal{F}}$ being closed. $\square$

Proposition 3.6. *The set $\hat{\nabla}$ is closed in $\hat{\mathcal{F}}$ in the Euclidean topology. In particular, it is closed in $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{\geq 0} \cap \Delta$.*

Proof. Let $\{(a_k, b_k, y_k)\}_{k \in \mathbb{N}} \subseteq \hat{\nabla}$ be a sequence converging to $(a, b, y) \in \hat{\mathcal{F}}$. At least one of the following two scenarios must occur: either there exists a proper face Γ of P such that $\hat{\mathcal{I}}_{>0}^\Gamma$ contains a subsequence, or $\hat{\nabla}^\circ$ contains a subsequence. If some $\hat{\mathcal{I}}_{>0}^\Gamma$ contains a subsequence $\{(a_{i_k}, b_{i_k}, y_{i_k})\}_{k \in \mathbb{N}}$, then by Proposition 3.5 (i) we have $(a, b, y) \in \hat{\mathcal{I}}_{>0}^{\Gamma'}$ for a face Γ' of Γ , so $(a, b, y) \in \hat{\nabla}$. Otherwise, if $\hat{\nabla}^\circ$ contains a subsequence $\{(a_{i_k}, b_{i_k}, y_{i_k})\}_{k \in \mathbb{N}}$, we have that $\det(J_{F_{a_{i_k}, b_{i_k}}}(\varphi_{\hat{M}}^{-1}(y_{i_k}))) = 0$ for $(a_{i_k}, b_{i_k}, y_{i_k}) \in \hat{\mathcal{I}}_{>0}$. By Proposition 3.5, either $(a, b, y) \in \hat{\mathcal{I}}_{>0}^\Gamma \subseteq \hat{\nabla}$

for some proper face Γ of P or $(a, b, y) \in \hat{\mathcal{I}}_{>0}$. In the latter case, as the determinant condition is a closed condition on the coordinates of $\hat{\mathcal{I}}_{>0}$, we get that $(a, b, y) \in \hat{\nabla}^\circ \subseteq \hat{\nabla}$. $\square$

3.3. Proof of the main result. Working in the toric setting pays off in the proof of [Theorem 2.4](#), where we crucially use that $(Y_{\hat{M}})_{\geq 0} \cap \Delta$ is compact.

Proof of Theorem 2.4. It suffices to show that the fiber of π is locally constant around any $(a, b) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \setminus \pi(\nabla_{\text{part}})$. By (iv) in [Proposition 3.5](#), $\pi(\mathcal{F})$ is closed and on its complement π has empty fiber. Hence, π is locally constant around $(a, b) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \setminus \pi(\mathcal{F})$. Consider $(a, b) \in \pi(\mathcal{F}) \setminus \pi(\nabla)$. By definition of ∇_{part} , $(a, b) \in \pi(\mathcal{I}_{>0}) \setminus \pi(\nabla^\circ)$. Hence, (a, b) is a regular value of both $\pi|_{\mathcal{I}_{>0}}: \mathcal{I}_{>0} \rightarrow \mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ and $\tau|_{\hat{\mathcal{I}}_{>0}}: \hat{\mathcal{I}}_{>0} \rightarrow \mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ by [Remark 3.4](#). By the diagram in (3.3), the fibers of $\pi|_{\mathcal{I}_{>0}}$ and $\tau|_{\hat{\mathcal{I}}_{>0}}$ are isomorphic. Since $\pi(\mathcal{I}_{>0}) \setminus \pi(\nabla^\circ) = \tau(\hat{\mathcal{I}}_{>0}) \setminus \tau(\hat{\nabla}^\circ)$, we are reduced to show the fiber of τ is locally constant around $(a, b) \in \tau(\hat{\mathcal{I}}_{>0}) \setminus \tau(\hat{\nabla}^\circ)$.

Since we are in the case $s + \ell = n$, $\hat{\mathcal{I}}_{>0}$ and $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ have the same dimension as smooth manifolds. It follows from the regular value theorem for smooth manifolds that $\tau^{-1}(a, b)$ is a discrete set of points. Since $\tau^{-1}(a, b)$ is a closed set in the compact $\{a\} \times \{b\} \times ((Y_{\hat{M}})_{\geq 0} \cap \Delta)$, it is also closed and hence finite. Write $\tau^{-1}(a, b) = \{(a, b, y_1), \dots, (a, b, y_d)\}$. By the implicit function theorem applied via the local coordinates of the manifolds, for each $i \in [d]$ there exists an open neighbourhood $U_i \subseteq \hat{\mathcal{I}}_{>0} \subseteq \hat{\mathcal{F}}$ of (a, b, y_i) and an open neighbourhood $V_i \subseteq \mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ of (a, b) such that $\tau|_{U_i}: U_i \rightarrow V_i$ is a homeomorphism. By possibly intersecting with the connected component, we can assume that each V_i is contained in it. Moreover, the sets can be chosen so that $U_i \cap U_j = \emptyset$ for $i \neq j$ and, after replacing each U_i by $\tau|_{U_i}^{-1}(V)$ where $V := \cap_{i=1}^d V_i$, we can assume that $\tau(U_i) = V$ for all $i \in [d]$. Consider now the set $\mathcal{W} := \hat{\mathcal{F}} \setminus (U_1 \cup \dots \cup U_d)$, which is closed in $\hat{\mathcal{F}}$ and also closed in $(Y_{\hat{M}})_{\geq 0} \cap \Delta$ because $\hat{\mathcal{F}}$ is closed in $(Y_{\hat{M}})_{\geq 0} \cap \Delta$ by [Proposition 3.5](#). As $\mathcal{W}$ is closed and τ is a closed map because $(Y_{\hat{M}})_{\geq 0} \cap \Delta$ is compact, $\tau(\mathcal{W})$ is closed in $\mathbb{R}_{>0}^m \times \mathbb{R}^\ell$. Hence, the set $V \setminus \tau(\mathcal{W}) \subset \mathbb{R}_{>0}^m \times \mathbb{R}^\ell$ is an open neighbourhood of (a, b) such that $\#\tau^{-1}(a', b') = d$ for all $(a', b') \in V \setminus \tau(\mathcal{W})$. $\square$

4. APPLICATIONS

4.1. Convex coordinates. It is clear from its definition that $\mathcal{I}_{>0} \neq \emptyset$ if and only if $\ker(C) \cap \mathbb{R}_{>0}^m \neq \emptyset$. The set $\ker(C) \cap \mathbb{R}_{>0}^m$ is a finitely generated open convex cone on $\mathbb{R}_{>0}^m$. Its set of extreme rays determine another parametrization of $\mathcal{I}_{>0}$ that sometimes is more suitable for computations. A matrix $E \in \mathbb{R}^{m \times r}$ whose columns are a choice of extreme vectors of the cone $\ker(C) \cap \mathbb{R}_{>0}^m$ defines the *convex* parametrization

$$\psi_{>0}: \mathbb{R}_{>0}^d \times \mathbb{R}_{>0}^n \rightarrow \mathcal{I}_{>0}; \quad (\lambda, h) \mapsto ((E\lambda) \star h^M, Lh^{-1}, h^{-1}). \quad (4.1)$$

Our strategy to identify connected components with constant topology consists of using the parametrizations of $\mathcal{I}_{>0}$ to define subsets that project down to discriminating sets on the parameter space. This is inspired by the definition and properties of the *critical polynomial* from reaction networks theory [1].

The set $\mathcal{I}_{>0} \cap \nabla$ discriminates the cardinality of the fibers of π inside $\mathcal{I}_{>0}$. Using the convex parametrization in (4.1) this set can be described as the positive zero set of a system of polynomial equations, which might be useful for computations.

Corollary 4.1. *Let $F \in \mathbb{R}[a, b, x^{\mathbb{R}}]^n$ be an augmented vertical system where C and M partition into r blocks as described in [Section 2.3](#). Assume $\text{rank}(C) = s$, $\text{rank}(L) = \ell$, $s + \ell = n$, and $\text{rank}(\hat{M}) = n + r$. Let $\psi_{>0}$ be the convex parametrization from (4.1) and let Σ_{part} be the set of*

multistationarity region would appear is that two of the 5 steady states collide and become a degenerate double root.

5. THE CASE WHEN $\text{conv}(M)$ IS NOT FULL DIMENSIONAL

(by Elisenda) If the system is purely vertical, then it will be degenerate and have generically no zeros. So this case only makes sense when we have linear relations L .

A first attempt could be to define $\hat{\mathcal{I}}_{>0}$ as

$$\hat{\mathcal{I}}_{>0} := \left\{ (a, b, y) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{>0} \cap \Delta : C(a \star y) = 0, Lx = b \text{ for some } x \in \varphi_{\hat{M}}^{-1}(y) \right\}.$$

We would still have a smooth map

$$\nu: \mathcal{I}_{>0} \longrightarrow \hat{\mathcal{I}}_{>0} \quad (a, b, x) \mapsto (a, b, \varphi_{\hat{M}}(x))$$

and a commutative diagram

$$\begin{array}{ccc} \mathcal{I}_{>0} & \xrightarrow{\nu} & \hat{\mathcal{I}}_{>0} \\ \pi \downarrow & & \downarrow \tau \\ \mathbb{R}_{>0}^m \times \mathbb{R}^\ell & \xrightarrow{\text{id}} & \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \end{array} \quad (5.1)$$

However, it is unclear to me how to work with ν . When $\text{rank}(\hat{M})$ is not maximal, then the fiber of $\varphi_{\hat{M}}$ at y is of the form

$$\rho_y = \{x \star t^A\}$$

for x any point in the fiber and A a matrix whose rows form a basis of the orthogonal complement of the direction of $\text{conv}(M)$. Call d the number of rows of A . This can for example be found by computing a basis of the left kernel of $\hat{M}$ and removing the first entry of each vector in the basis.

It is not obvious what the set of points $z \in \rho_y$ that satisfy $Lz = b$ is. It is likely that for nondegenerate systems, this set is finite, but how the cardinality behaves with respect to the parameters a, b is not clear to me. It is not straightforward to understand the dimension of $\hat{\mathcal{I}}_{>0}$.

An alternative construction that might be better behave is as follows. Choose a smooth section σ of $\varphi_{\hat{M}}$, which is a way of picking a representative in each fiber (to make precise how to). Then, given $y = \varphi_{\hat{M}}(x) \in (Y_{\hat{M}})_{>0} \cap \Delta$, we can consider the point $\sigma(y) \in \mathbb{R}_{>0}^n$, and if z is another point in the fiber of y , then $z/\sigma(y) = t^A$ for some t .

Consider the positive toric variety $(Y_A)_{>0}$ obtained from the matrix A above. Then a still wrong definition could be

$$\hat{\mathcal{I}}_{>0} := \left\{ (a, b, y, \delta) \in \mathbb{R}_{>0}^m \times \mathbb{R}^\ell \times (Y_{\hat{M}})_{>0} \cap \Delta \times (Y_A)_{>0} : C(a \star y) = 0, L(\sigma(y) \star \delta) = b \right\}.$$

Then for any $(a, b, x) \in \mathcal{I}_{>0}$, we consider the map

$$(a, b, x) \mapsto (a, b, \varphi_{\hat{M}}(x), x/\sigma(\varphi_{\hat{M}}(x))) \in \hat{\mathcal{I}}_{>0}$$

with inverse

$$(a, b, y, \delta) \mapsto (a, b, \sigma(y) \star \delta) \in \mathcal{I}_{>0}.$$

This is still unsatisfactory, because $(Y_A)_{>0}$ is not compact. But maybe we could do the same trick as with M , and consider $(Y_{\hat{A}})_{\geq 0} \cap \Delta$ instead. One should work out the details of whether this set still explains the fiber of $\varphi_{\hat{M}}$.

If it does, then we have an isomorphism of incidence varieties, and maybe the results carry over, potentially after considering the faces of $\text{conv}(A)$ as well. That wouldn't be that nice, because A depends on our choice of orthogonal complement, and although it can be made to have integer entries, the polytope can change quite a lot. But in the end this is for the proof, so we may not need to compute anything, so we do not care.

6. THINGS TO ADDRESS

Thoughts on how to handle the case where the kernel of C has a partition, by EF:

We have the problem that the main theorem is silent if the kernel of C partitions in such a way that any vector in the kernel of C is in the kernel of the restriction to some face. In this case, ∇ is everything. This happens for the phosphorylation networks.

It could be that this can be fixed by incorporating the partitions into the set up, as we do with Oskar and also with Murad (and Stefan and Georg do). Let's say that C in row reduced form is block diagonal, with blocks C_1, C_2 , such that $\ker(C) = \ker(C_1) \times \ker(C_2)$. Let $\mathcal{M}_1, \mathcal{M}_2$ be the exponent vectors corresponding to the columns of C_1, C_2 respectively. Then it can well be that we can consider the faces of $\text{conv}(\mathcal{M}_i)$, $i = 1, 2$, instead of the faces of $\text{conv}(\mathcal{M})$. This would probably mean that the toric variety needs to be obtained from the matrix $\widehat{M}$ with two extra rows on top, one with 1's on top of the elements of $\mathcal{M}_1$ and the other for $\mathcal{M}_2$. Then the compactification would be with two simplices, one for each part: $Y \cap (\Delta_{m_1-1} \times \Delta_{m_2-1})$.

In the proofs, the important fact would probably be that $v \in \ker(C)$ if and only if its two components v_1, v_2 satisfy $v_i \in \ker(C_i)$. Then it can be that all details, like Prop 3.2, carry through.

So, my hope is that a ∇ defined with ∇^0 as currently is, and the kernel of the restriction of C_i to proper faces of $\text{conv}(\mathcal{M}_i)$, works.

Of course this should work with more partitions if what I say holds.

MT: I like the idea, I think this should work. Is this the only situation when ∇ is everything?

Thoughts on computational aspects, by EF: If the corollary is rephrased to address the regions where the number of zeros is constant, what we have is a two step approach for understanding the possible number of zeros:

- First, you study the linear part, consider only the faces leading to a hyperplane, and then the complements divide the parameter regions into cones in the λ 's. As the λ 's are positive, these are cones in the positive orthant. Now, we can parametrize each cone separately by taking its extreme rays, so the parameters again take positive values.
- For each cone in the λ 's, we plug its parametrization into the critical polynomial, and study the signs. For some cones, it might be that the polynomial has a constant sign, so in that cone the number of zeros is constant. By choosing a λ there and any h , we find the number of zeros. For other cones, there might be several regions. These can be studied with separating hyperplanes and other methods, and we can explore whether there are other number of zeros. A way to approach this would be to take a ray in the cone and see how it intersects the zero set of the critical polynomial.

It would be worth playing with this for vertical systems that are pure and without partitions of the kernel of C . For example, for the Viro critical systems for separable supports.

The discriminant is the whole parameter space MT: For some parametric families, it might happen that the projection of ∇ onto the parameter space is the whole parameter space. Here is an example.

Example 6.1. Consider the vertical system

$$\begin{aligned} a_1 x^2 + a_2 y^2 - a_3 &= 0 \\ x - y &= b \end{aligned}$$

The coefficient and exponent matrix equals

$$C = \begin{pmatrix} 1 & 1 & -1 \end{pmatrix}, \quad M = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \end{pmatrix}$$

Looking at the face $\Gamma = \{(2, 0), (0, 0)\}$, we have

$$C^\Gamma = \begin{pmatrix} 1 & 0 & -1 \end{pmatrix}$$

Note that the projection of $\mathcal{I}_{>0}^\Gamma$ onto $\mathbb{R}_{>0}^3 \times \mathbb{R}^1$ is surjective.

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